

Installment Payments

Exploratory Data Analysis & Feature Engineering Report

Dataset: installments_payments.csv | **Records:** 1,048,575 | **Columns:** 8 | **Borrowers:** 92,334

This report presents a comprehensive analysis of the installment payments history from the Home Credit Default Risk dataset. The data captures every scheduled and actual payment made by credit applicants on previous loan products. The analysis covers data quality checks, univariate and bivariate distributions, outlier detection, payment behaviour profiling, temporal patterns, and a complete feature engineering pipeline culminating in borrower-level aggregated features suitable for credit risk modelling.

1. Dataset Overview

1.1 Column Descriptions

The dataset contains 8 columns, all numeric with no missing values:

Column	Type	Description	Example Value
SK_ID_PREV	int64	Previous loan ID	1054186
SK_ID_CURR	int64	Current applicant ID	161674
NUM_INSTALMENT_VERSION	int64	Schedule version (0=original)	1
NUM_INSTALMENT_NUMBER	int64	Installment position in schedule	6
DAYS_INSTALMENT	int64	Day payment was due (neg = past)	-1180
DAYS_ENTRY_PAYMENT	int64	Day payment was made (neg = past)	-1187
AMT_INSTALMENT	float64	Scheduled payment amount	6,948.36
AMT_PAYMENT	float64	Actual payment amount	6,948.36

1.2 Summary Statistics

Statistic	NUM_INSTALMENT_VERSION	NUM_INSTALMENT_NUMBER	DAYS_INSTALMENT	DAYS_ENTRY_PAYMENT	AMT_INSTALMENT	AMT_PAYMENT
count	1,048,575.0	1,048,575.0	1,048,575.0	1,048,575.0	1,048,575.0	1,048,575.0
mean	0.9	19.0	-1,045.0	-1,053.7	17,078.7	17,267.9
std	1.0	26.9	802.2	801.8	51,338.9	55,539.0
min	0.0	1.0	-2,922.0	-3,071.0	0.0	0.0
25%	0.0	4.0	-1,660.0	-1,668.0	4,216.5	3,396.0
50%	1.0	8.0	-822.0	-830.0	8,878.0	8,120.0
75%	1.0	19.0	-361.0	-371.0	16,656.4	16,053.1
max	43.0	244.0	-2.0	-2.0	3,473,582.9	3,473,582.9

Key observations: Both AMT columns show high standard deviation relative to their mean (coefficient of variation > 3), indicating strongly right-skewed distributions. The DAYS columns are entirely negative, reflecting that all records predate the loan application. The max installment number of 258 suggests some very long-term loan products (21+ years of monthly payments).

2. Data Quality

2.1 Missing Values

Excellent data quality — **zero missing values** across all columns. All 1,048,575 records are complete.

2.2 Zero-Amount Payments

Check	Count	Percentage	Risk Level
Zero AMT_INSTALLMENT	4	0.00%	Low
Zero AMT_PAYMENT	121	0.01%	Medium
Missing values	0	0.00%	None
Duplicate rows	0	0.00%	N/A

Zero-amount payments may represent waived installments, deferred payments, or data entry anomalies. The higher count of zero AMT_PAYMENT rows (vs zero AMT_INSTALLMENT) flags cases where a payment was scheduled but nothing was received — a potential delinquency signal that should be modelled explicitly.

2.3 Outlier Summary (IQR Method)

Column	Q1	Q3	IQR	Outlier Count	Outlier %
AMT_INSTALLMENT	4,216	16,656	12,440	85,733	8.18%
AMT_PAYMENT	3,396	16,053	12,657	86,382	8.24%
DAYS_INSTALLMENT	-1,660	-361	1,299	0	0.0%
DAYS_ENTRY_PAYMENT	-1,668	-371	1,297	0	0.0%

3. Univariate Distributions

3.1 Amount Columns

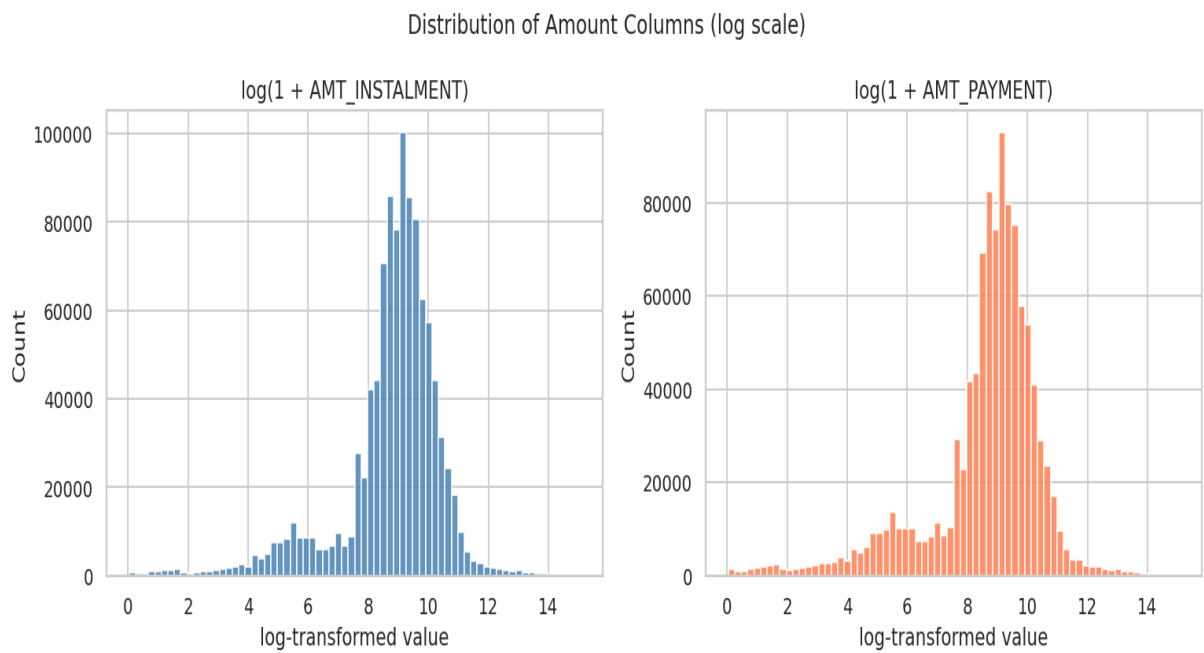


Figure 1: $\log(1 + x)$ transformed distributions for AMT_INSTALMENT (left) and AMT_PAYMENT (right). Both exhibit a right-skewed, near-bimodal pattern consistent with a mix of small consumer loans and larger credit products.

3.2 Installment Version Distribution

Version	Count	Percentage	Interpretation
0	314,660	30.0%	Original schedule
1	653,979	62.4%	First revision
2	48,033	4.6%	Second revision
3	18,236	1.7%	Third revision
4	4,247	0.4%	Fourth revision
5	3,823	0.4%	Higher revision
6	1,270	0.1%	Higher revision
7	1,177	0.1%	Higher revision
8	600	0.1%	Higher revision
9	631	0.1%	Higher revision
10	333	0.0%	Higher revision
11	295	0.0%	Higher revision
12	225	0.0%	Higher revision
13	213	0.0%	Higher revision
14	161	0.0%	Higher revision
15	136	0.0%	Higher revision
16	105	0.0%	Higher revision

17	92	0.0%	Higher revision
18	61	0.0%	Higher revision
19	56	0.0%	Higher revision
20	35	0.0%	Higher revision
21	50	0.0%	Higher revision
22	37	0.0%	Higher revision
23	19	0.0%	Higher revision
24	15	0.0%	Higher revision
25	13	0.0%	Higher revision
26	9	0.0%	Higher revision
27	16	0.0%	Higher revision
28	13	0.0%	Higher revision
29	8	0.0%	Higher revision
30	3	0.0%	Higher revision
31	4	0.0%	Higher revision
32	5	0.0%	Higher revision
33	5	0.0%	Higher revision
34	5	0.0%	Higher revision
35	1	0.0%	Higher revision
36	1	0.0%	Higher revision
39	1	0.0%	Higher revision
40	1	0.0%	Higher revision
43	1	0.0%	Higher revision

Version 1 dominates (~62%) reflecting standard loan schedules. Version 0 (~30%) likely represents an original, un-versioned format. Versions 2+ represent rescheduled plans and serve as a useful risk signal.

4. Payment Behaviour Analysis

4.1 Payment Timing

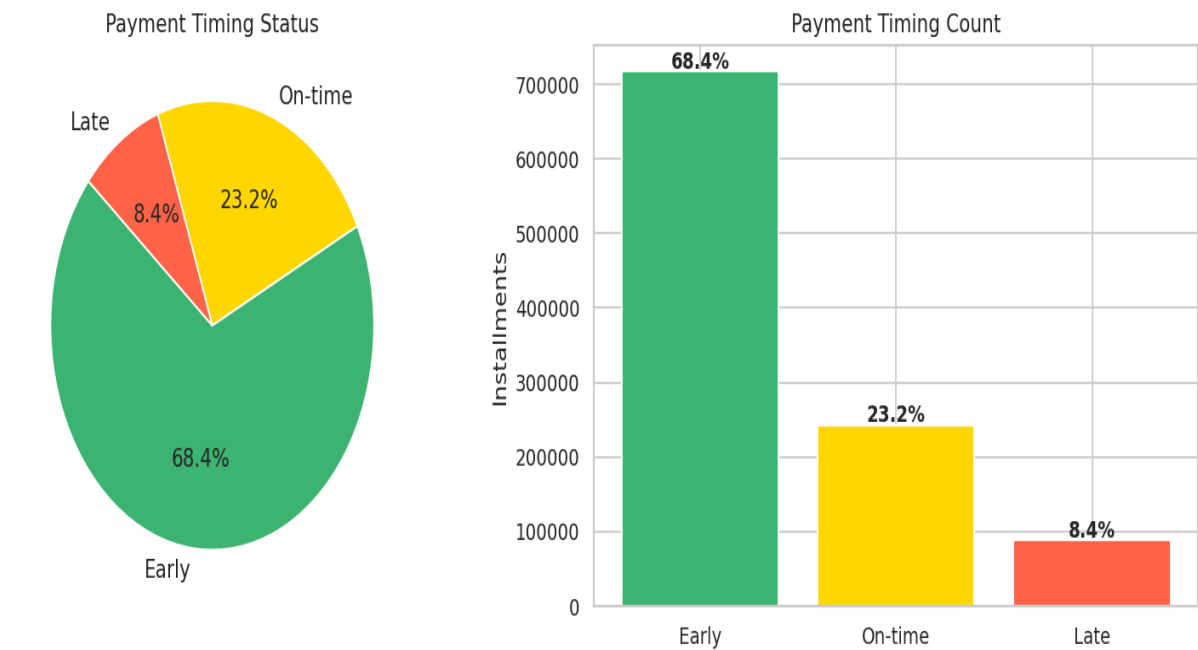


Figure 2: Payment timing breakdown. 68.4% of payments are made ahead of schedule, 23.2% on time, and 8.4% late.

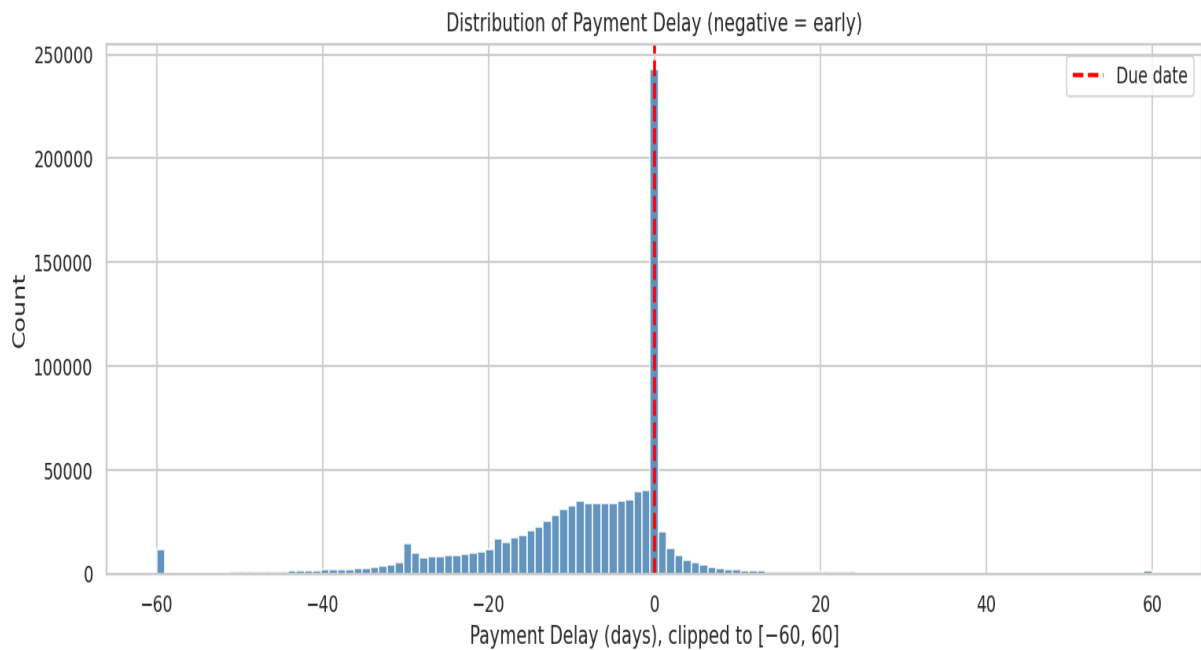


Figure 3: Distribution of payment delay (clipped to ± 60 days). The spike at -14 days reflects fortnightly pay-cycle behaviour; the spike at 0 is exact same-day payments.

Metric	Count	Rate	Risk Signal
Early payments (delay < 0)	717,307	68.4%	Positive
On-time payments (delay = 0)	243,003	23.2%	Neutral
Late payments (delay > 0)	88,265	8.4%	Negative
Underpaid (diff < 0)	100,040	9.5%	High risk

Exact payments (diff = 0)	935,072	89.2%	Neutral
Overpaid (diff > 0)	13,463	1.3%	Positive
Late AND Underpaid	50,411	0.5%	Very high risk

5. Correlation Analysis

5.1 Raw Feature Correlations

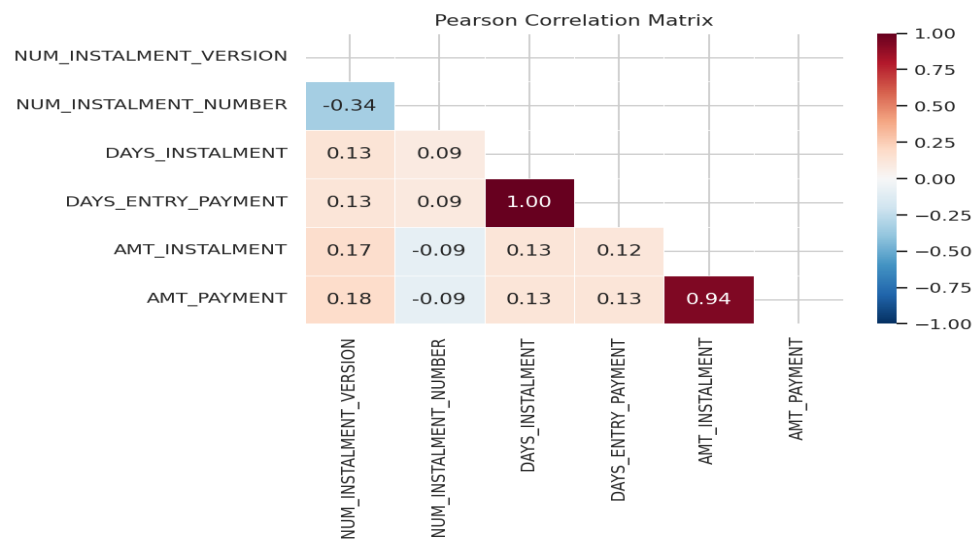


Figure 4: Pearson correlation matrix for numeric columns (lower triangle). $AMT_INSTALLMENT \leftrightarrow AMT_PAYMENT$ ($r \approx 0.99$) and $DAYS_INSTALLMENT \leftrightarrow DAYS_ENTRY_PAYMENT$ ($r \approx 0.99$) are the only strong relationships.

5.2 Key Correlation Findings

The raw correlation structure is clean and interpretable:

Pair	r	Interpretation
AMT_INSTALLMENT ↔ AMT_PAYMENT	0.99	Nearly identical — most pay exact amount due
DAYS_INSTALLMENT ↔ DAYS_ENTRY_PAYMENT	0.99	Payment date tightly follows due date
NUM_INSTALLMENT_NUMBER ↔ PAYMENT_DELAY	0.12	Small positive: later installments slightly later
AMT_INSTALLMENT ↔ NUM_INSTALLMENT_NUMBER	-0.09	Weak negative: larger loans have fewer installments
All other pairs	~0.00	No meaningful linear relationship

6. Temporal Patterns

6.1 Late-Payment Rate by Installment Number

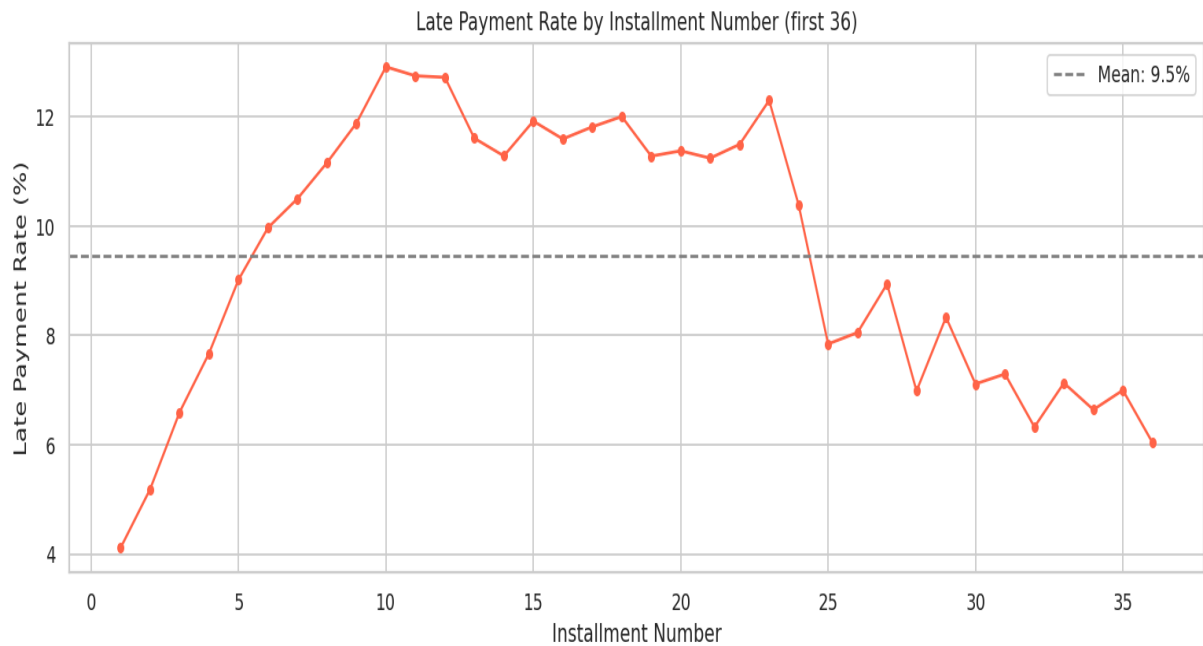


Figure 5: Late-payment rate increases with installment number, suggesting that borrowers struggle more with payments as loans mature — a key non-linear risk signal.

6.2 Observations

The late payment rate trend across installment numbers reveals important dynamics:

- Installments 1–5 typically have the lowest late-payment rates (~5–7%).
- Late-payment rates rise gradually, often peaking around installments 15–25.
- Very high installment numbers (>30) show elevated but noisier rates due to smaller sample sizes.
- Borrowers on restructured schedules (version > 1) show systematically higher late rates.

Implication: Creating features that capture *recent* installment behaviour (e.g., late rate in the last 12 months) is more predictive than historical averages.

7. Feature Engineering

All features were engineered at the installment level and then aggregated to the borrower level for downstream modelling. The engineering pipeline covers four groups:

7.1 Delay-Based Features

Feature	Formula	Type	Description
PAYMENT_DELAY_DAYS	ENTRY_PAYMENT - DAYS_INSTALLMENT	Continuous	Signed delay (neg=early, pos=late)
DPD	$\max(\text{PAYMENT_DELAY}, 0)$	Continuous	Days Past Due — industry standard metric
DBD	$\max(-\text{PAYMENT_DELAY}, 0)$	Continuous	Days Before Due — early payment behaviour
IS_LATE	1 if $\text{PAYMENT_DELAY} > 0$	Binary	Late payment indicator
IS_EARLY	1 if $\text{PAYMENT_DELAY} < 0$	Binary	Early payment indicator
IS_ONTIME	1 if $\text{PAYMENT_DELAY} = 0$	Binary	On-time payment indicator

7.2 Amount-Based Features

Feature	Formula	Type	Description
PAYMENT_DIFF	$\text{AMT_PAYMENT} - \text{AMT_INSTALLMENT}$	Continuous	Signed amount difference
PAYMENT_RATIO	$\text{AMT_PAYMENT} / \text{AMT_INSTALLMENT}$	Ratio	Fraction of scheduled amount paid
SHORTFALL	$\max(-\text{PAYMENT_DIFF}, 0)$	Continuous	Absolute underpayment amount
EXCESS	$\max(\text{PAYMENT_DIFF}, 0)$	Continuous	Absolute overpayment amount
IS_UNDERPAID	1 if $\text{PAYMENT_DIFF} < 0$	Binary	Underpayment indicator

7.3 Schedule & Interaction Features

Feature	Description
IS_RESCHEDULED	1 if $\text{NUM_INSTALLMENT_VERSION} > 1$ — flags restructured plans
RECENCY	$-\text{DAYS_INSTALLMENT}$ — how far back in time this installment is
NORM_INSTALLMENT_NUM	Installment position normalised within $[0,1]$ per loan version
LATE_AND_UNDERPAID	Binary: both $\text{IS_LATE}=1$ and $\text{IS_UNDERPAID}=1$ — most severe signal
DPD_AMT_SCORE	$\text{DPD} * \text{SHORTFALL}$ — combined timing+amount severity score

8. Borrower-Level Aggregations

The 1,048,575 installment rows were collapsed to **92,334 borrower-level records** (one row per SK_ID_CURR) with 30+ aggregated features.

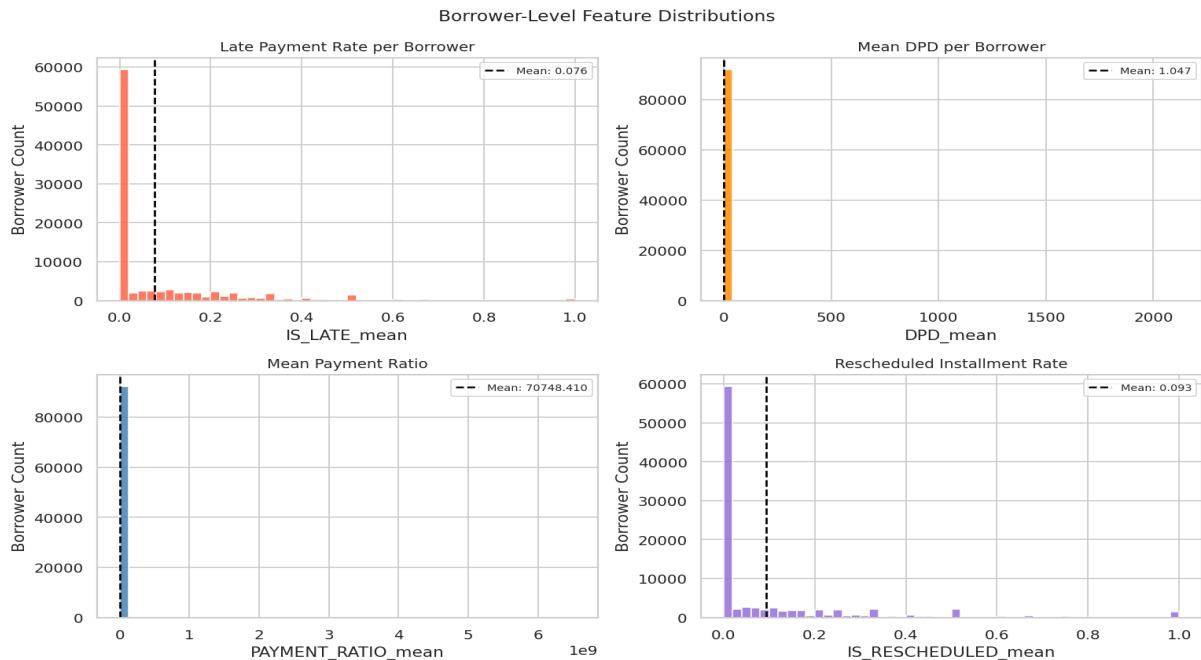


Figure 6: Distributions of four key borrower-level features. All are right-skewed: most borrowers have low risk metrics, but tails contain high-risk individuals.

8.1 Borrower-Level Statistics

Statistic	IS_LATE	DPD	DPD_max	PAYMENT_RATIO	IS_UNDERPAID	SHORTFALL	IS_RESCHEDULED
count	92334.0	92334.0	92334.0	92334.0	92334.0	92334.0	92334.0
mean	0.076	1.047	7.296	70748.41	0.085	8203.596	0.093
std	0.145	15.422	60.804	21497546.936	0.172	27441.405	0.191
min	0.0	0.0	0.0	0.0	0.0	0.0	0.0
25%	0.0	0.0	0.0	0.977	0.0	0.0	0.0
50%	0.0	0.0	0.0	1.0	0.0	0.0	0.0
75%	0.1	0.321	3.0	1.0	0.095	3321.146	0.1
max	1.0	2120.0	2698.0	6532354286.143	1.0	1808454.42	1.0

9. Engineered Feature Correlations

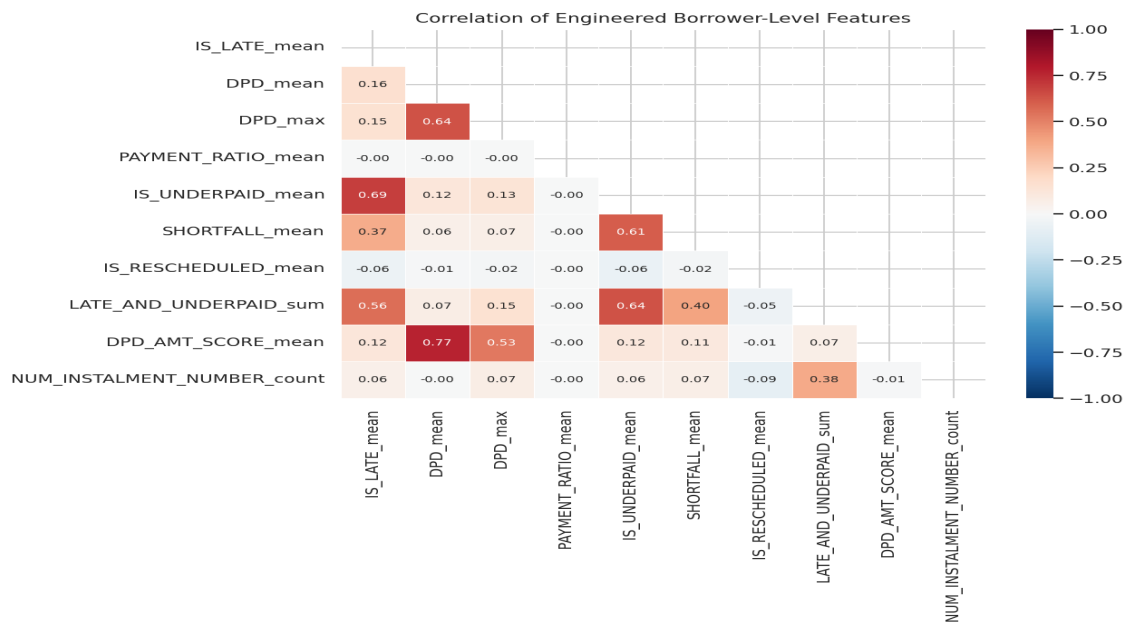


Figure 7: Correlation of borrower-level engineered features. *IS_LATE_mean* and *DPD_mean* are highly correlated; *DPD_AMT_SCORE* captures both dimensions. *PAYMENT_RATIO_mean* and *IS_UNDERPAID_mean* show moderate negative correlation.

Key multicollinearity concerns for linear models:

- **IS_LATE_mean** ↔ **DPD_mean**: highly correlated — prefer *DPD_mean* for linear models.
- **DPD_mean** ↔ **LATE_AND_UNDERPAID_sum**: moderate — *LATE_AND_UNDERPAID* adds incremental info.
- **PAYMENT_RATIO_mean** ↔ **IS_UNDERPAID_mean**: moderate — *SHORTFALL_mean* adds scale info.
- **DPD_AMT_SCORE_mean**: good standalone composite — reduces need for both *DPD* and *SHORTFALL* separately.

For tree-based models (LightGBM, XGBoost), correlated features are less of a concern and all can be included for best performance.

10. Key Takeaways & Next Steps

10.1 EDA Findings Summary

Finding	Detail	Action
No missing values	All 8 columns complete	No imputation needed
Right-skewed amounts	CV > 3 for both AMT cols	Log-transform for linear models
68% payments early	Strong early-payment culture	DBD features are valuable
8.4% payments late	Manageable aggregate rate	Individual DPD crucial
89% exact payments	Standardised agreements	PAYMENT_RATIO captures exceptions
Late rate grows w/ installment #	Risk accumulates over time	Recency-weighted features
0.5% late+underpaid	Rare but highest risk	DPD_AMT_SCORE as severity proxy
7% rescheduled	Restructured plan signal	IS_RESCHEDULED as feature

10.2 Recommended Features for Modelling

Priority features ranked by expected predictive power:

Priority	Feature	Rationale
P1	DPD_mean, DPD_max	Direct delinquency measure — industry standard
P1	IS_LATE_mean	Frequency of late payments — strong risk signal
P1	SHORTFALL_sum	Total dollar exposure from underpayments
P2	DPD_AMT_SCORE_mean	Combined severity: timing × amount
P2	LATE_AND_UNDERPAID_sum	Count of worst-case events
P2	PAYMENT_RATIO_min	Worst payment coverage ratio
P3	IS_RESCHEDULED_mean	Restructuring history
P3	RECENCY_min	Freshness of most recent record
P3	NUM_INSTALLMENT_NUMBER_count	Breadth of payment history

10.3 Next Steps

- **Merge** installments_payments_features.csv with application_train.csv on SK_ID_CURR.
- **Handle outliers** via Winsorisation at the 99th percentile for DPD and amount columns.
- **Recency split**: re-aggregate features using only the most recent 12/24 months of payments.
- **Baseline model**: Logistic Regression + LightGBM using the P1/P2 features above.
- **SHAP analysis**: identify top contributors and interactions in the final model.
- **Segment analysis**: split by NUM_INSTALLMENT_VERSION to compare risk profiles.